UGA
UFR IM²AG Université Grenoble Alpes

M1 Informatique / MOSIG



Introduction to modeling and verification of digital systems

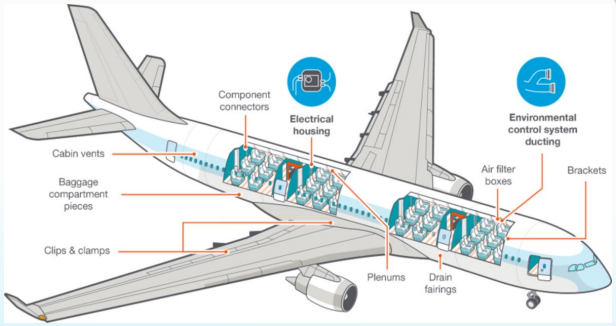
Laurence PIERRE
Laurence.Pierre@univ-grenoble-alpes.fr

TIMA

Rationale

- In embedded systems...

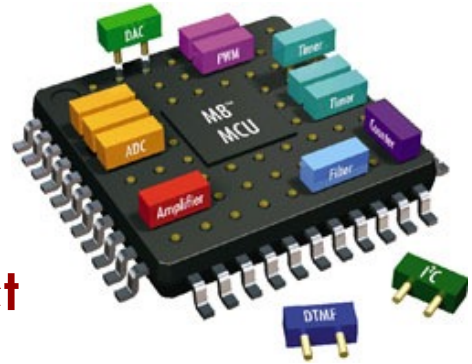








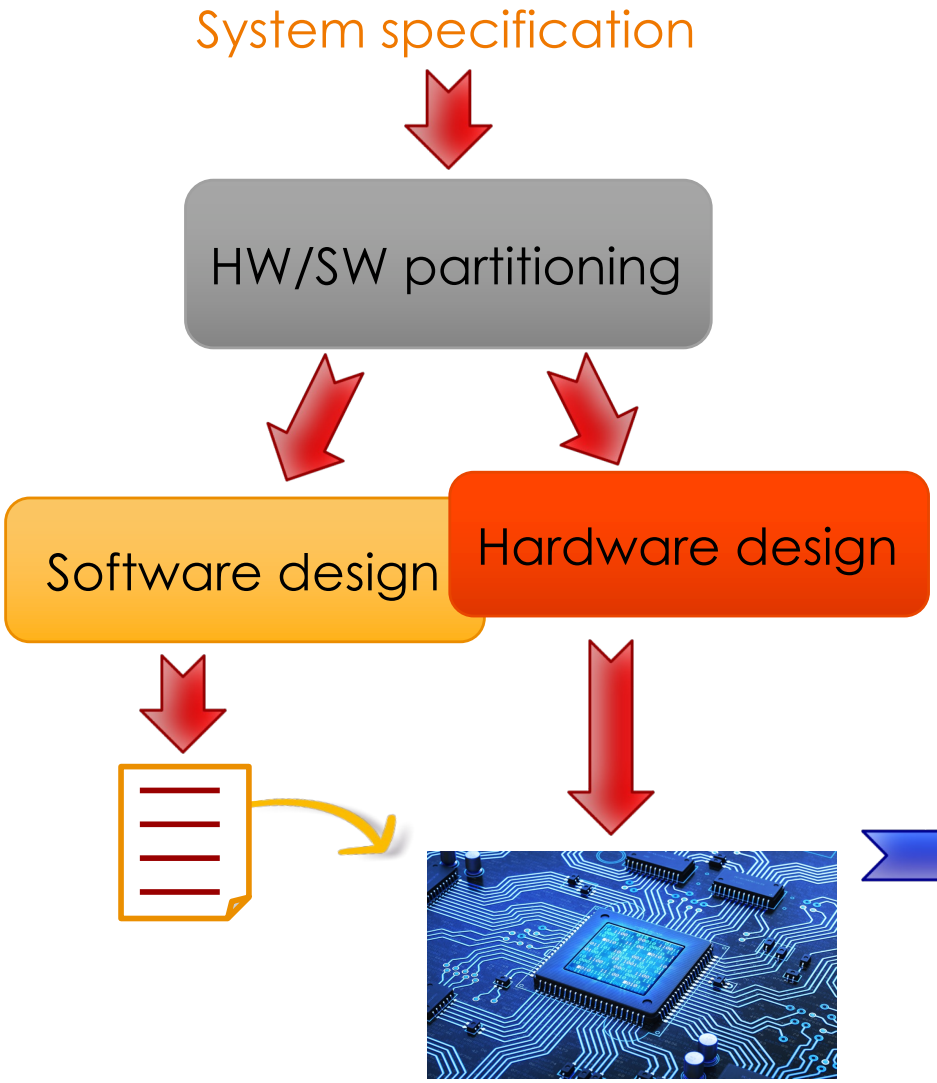


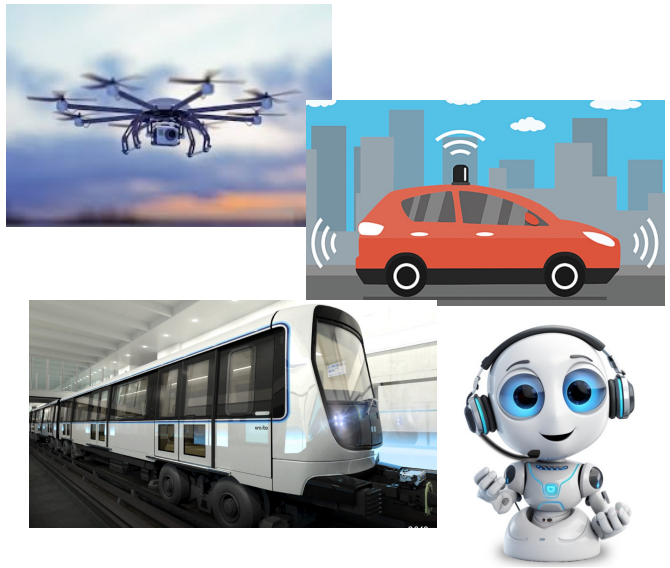


Hardware and **software** which **closely interact**

Rationale

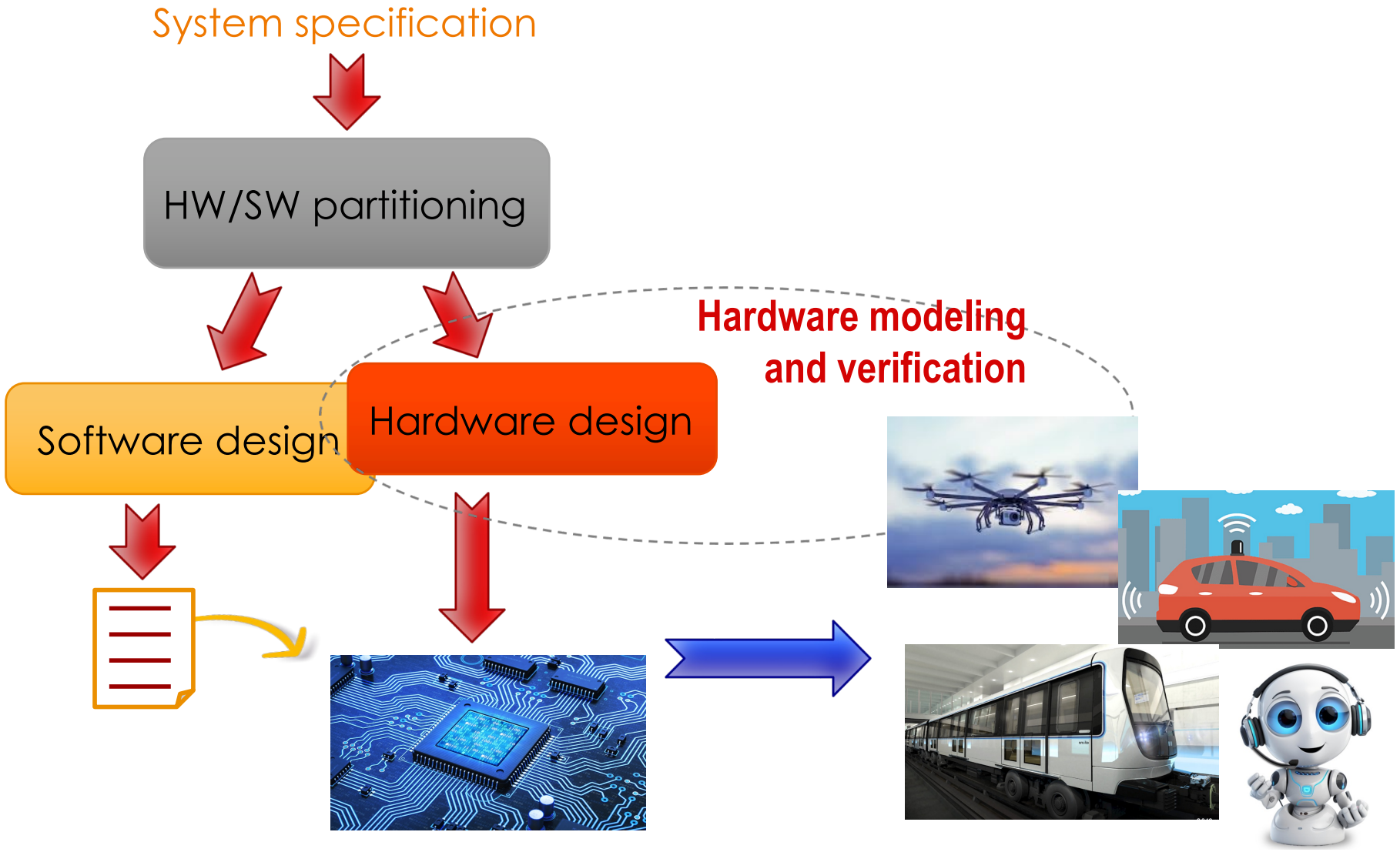
- **Embedded system design**





Rationale

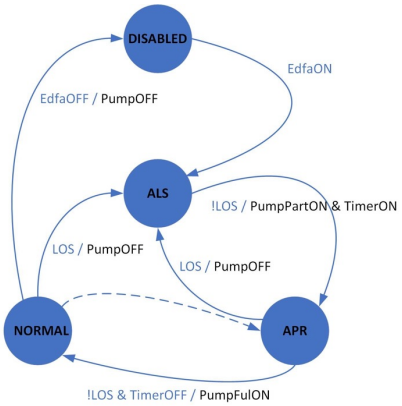
- **Embedded system design**



Hardware design ?

In this course

Specification of the expected behaviour

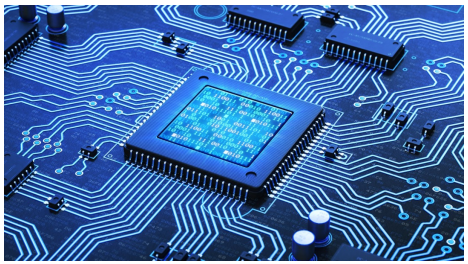




Modeling?

Verification?

Synthesis?

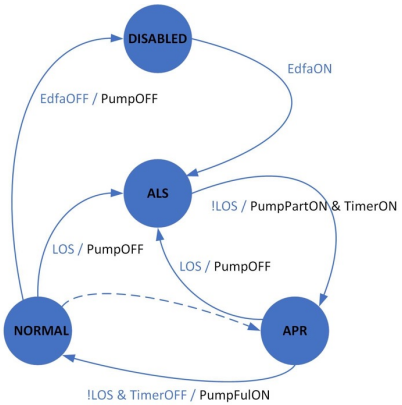


Hardware implementation

Hardware design ?

In this course

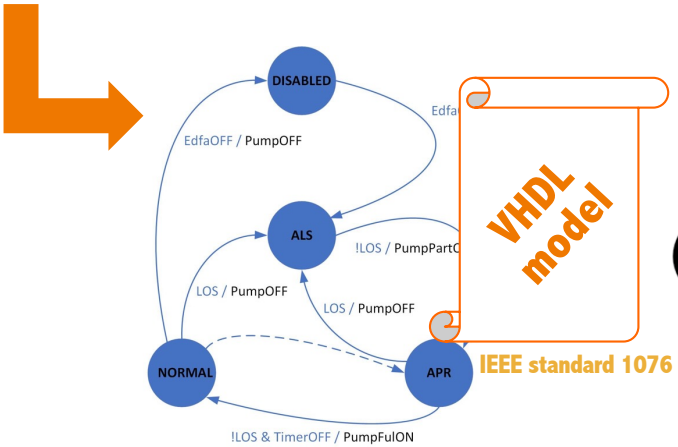
Specification of the expected behaviour



Hardware design ?

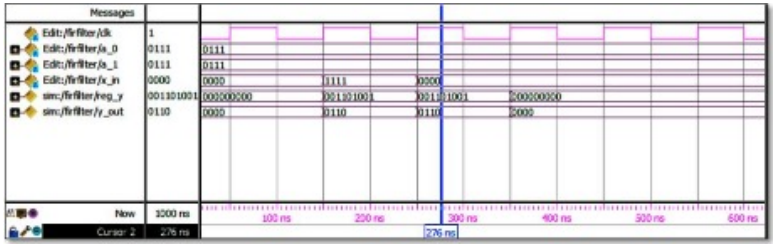
In this course

Specification of the expected behaviour



1

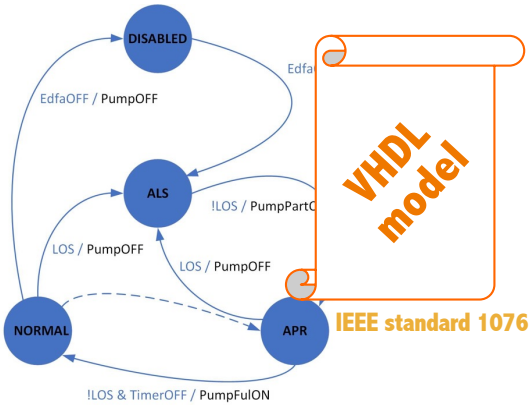
Simulation



Hardware design ?

In this course

Specification of the expected behaviour



Simulation

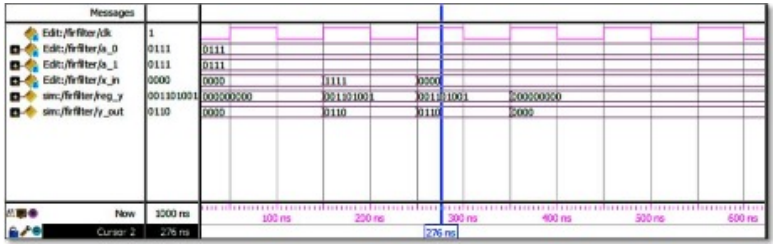
Correctness properties (temporal assertions)



$(\neg b)\ \mathsf{U}\ (a \wedge b)$

$\Box(a \rightarrow (\bigcirc \neg a \vee b))$

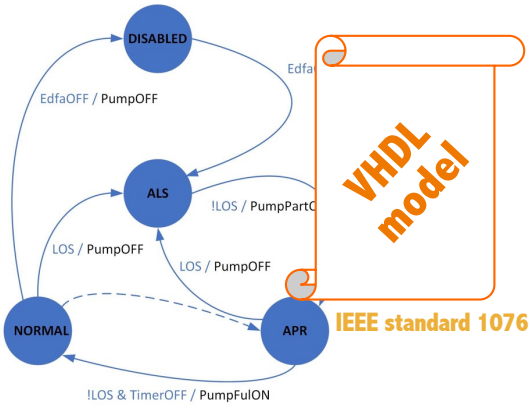
2



Hardware design ?

In this course

Specification of the expected behaviour



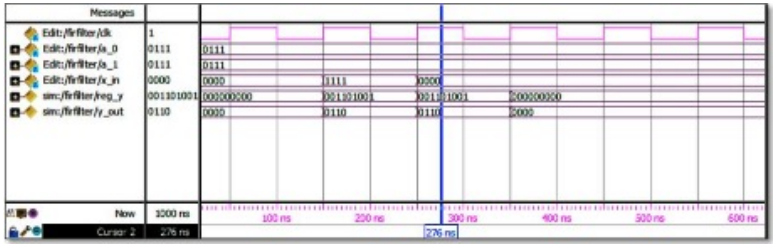
Correctness properties (temporal assertions)



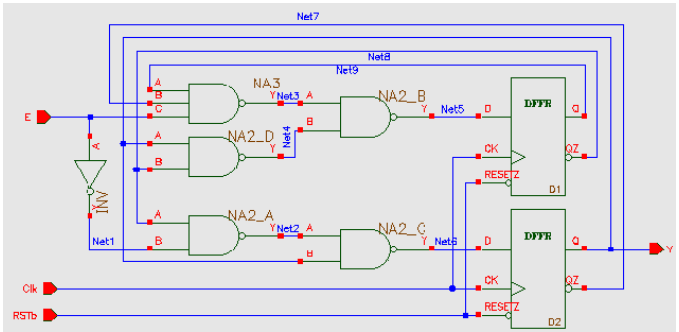
$(\neg b)\ \mathsf{U}\ (a \wedge b)$

$\square(a \rightarrow (\bigcirc \neg a \vee b))$

Simulation



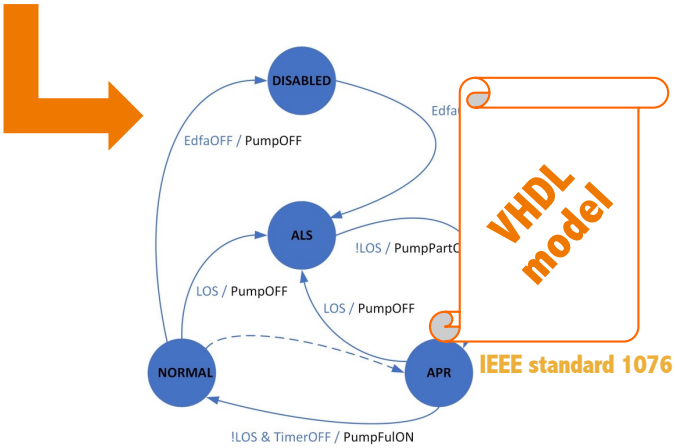
3 Logic **synthesis** (ASIC)



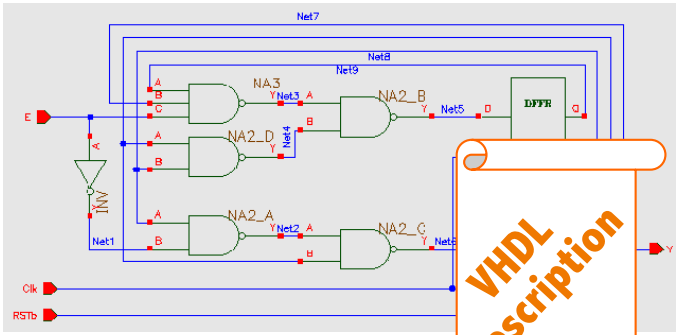
Hardware design ?

In this course

Specification of the expected behaviour



Logic **synthesis** (ASIC)



Simulation

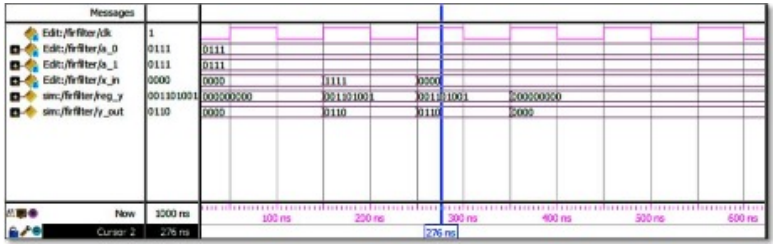
4

Correctness properties (temporal assertions)



$(\neg b)\ U\ (a \wedge b)$

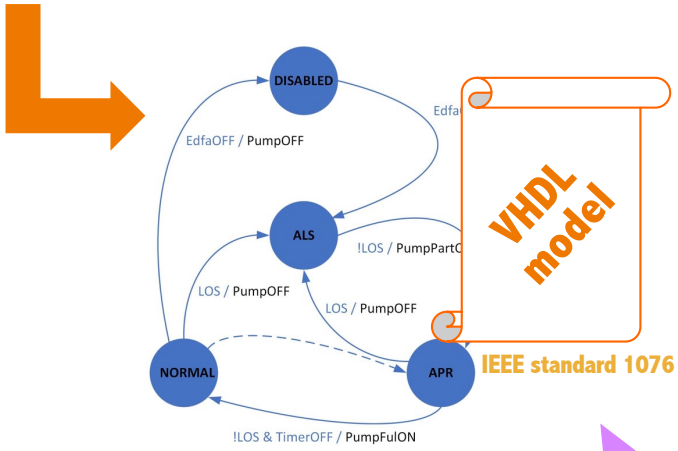
$\square(a \rightarrow (\bigcirc \neg a \vee b))$



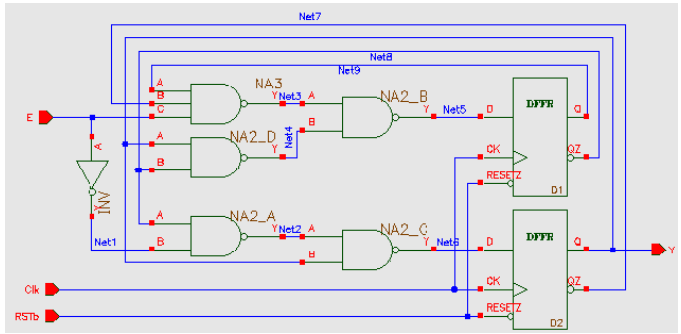
Hardware design ?

In this course

Specification of the expected behaviour



Logic **synthesis** (**ASIC**)



FPGA synthesis

5

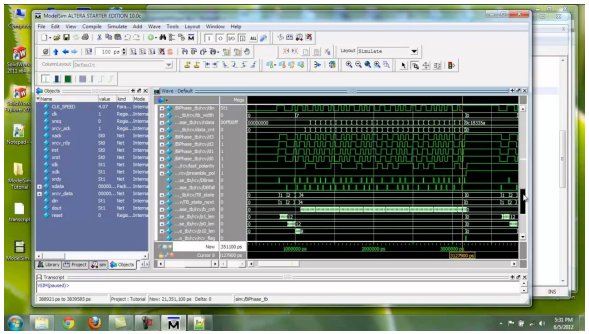


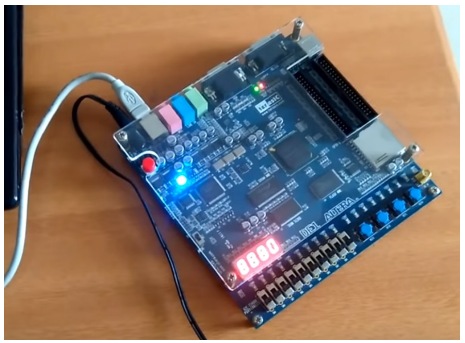
Hardware implementations

(with **hardware constraints**: frequency, area, consumption,...)

Organization

- **11 sessions**
  - 6 lecture+exercises sessions
  - 5 lab sessions
- **Prerequisite:** almost none (logic gates)
- **Assessment**
  - Complete project (5 lab sessions): from specification to FPGA, 40%





  - Exam, 60%

Organization

- Lab sessions at **CIME Nanotech** (Centre interuniversitaire de microélectronique et nanotechnologies)
  - Use of **industrial software tools**

